

Nicolas D. Swank

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SKILLS

JavaScript, Java, Node.js, Express.js, SQL Server, MongoDB, Mongoose, React, Postman, Jest, REST APIs, HTML, CSS, GIT, UML

EDUCATION

Eastern Michigan University

Bachelor of Science, Computer Science

Dean's List

Ypsilanti, Michigan

Cumulative GPA: 3.79

April 2024

PROJECTS

Data Warehouse & Analytics Project

SQL & Data Warehousing

- Built a data warehouse to transform raw source data (CSV files) into business ready tables for analysis.
- Designed bronze, silver, and gold layers to organize raw, cleaned, and aggregated data.
- Practiced ETL processes: loading, cleaning, and combining multiple data sources into a single analytical model.
- **Tech Stack:** SQL Server, T-SQL, CSV datasets, SSMS

To-do Task API - A backend RESTful API for managing personal tasks

Backend Web Development

- Built a backend API that allows users to sign up, log in, and manage to-do tasks.
- Organized route handling, authentication, and file uploads using Express routers and middleware.
- Implemented JWT-based authentication to ensure users can only access and modify their own tasks.
- Wrote automated tests using Jest and simulated frontend client requests using Postman.
- **Tech Stack:** Node.js, Express.js, MongoDB, Mongoose, JWT, Jest, Postman

Pathfinding Visualizer using OpenStreetMap

Algorithm Design Analysis

- Developed a web application in a team of 6 to determine distances between points on a map using Dijkstra's, A-Star, Breadth First Search, and Depth First Search algorithms.
- Integrated OpenStreetMap API to generate and display city maps dynamically.
- Implemented Dijkstra's algorithm using a priority queue to efficiently calculate shortest paths.
- **Tech Stack:** JavaScript, React, Node.js, OpenStreetMap API

Forkify - A recipe searching web application

Frontend Web Development

- Built a single-page recipe application to search, view, bookmark, and upload recipes using a public API.
- Structured the application using the MVC architecture to manage data, UI and business logic cleanly.
- Implemented a custom Publisher-Subscriber pattern to handle UI events such as searching, pagination, and uploading recipes in a modular and scalable way.
- **Tech Stack:** JavaScript, HTML/CSS, Parcel, API Integration

WORK EXPERIENCE

The Salvation Army - Adult Rehabilitation Center

Detroit, Michigan

IT Technician

Dec 2024 - Present

- Provide technical support via phone, email, and in-person interactions, troubleshooting hardware, software, and network issues across 40 locations.
- Install, configure, and troubleshoot hardware, LANs, WANs, and phone systems (mobile, VOIP, landline) as needed.
- Document support interactions in a ticketing system, ensuring detailed records of issues and resolutions.

Everwise Credit Union

South Bend, Indiana

ATM Digital Services Intern

Aug2023 - Dec 2024

- Resolved online banking platform and ATM issues through ServiceNow, enhancing user experience and platform reliability by troubleshooting system bugs, facilitating vendor escalations, and coordinating ATM tech dispatches.
- Utilized analytical and communication skills to diagnose and address online banking platform challenges, collaborating with teams and vendors to ensure operational efficiency and customer satisfaction.

IT Service Desk Intern

May 2023 - Aug 2023

- Assist end-users with incident resolution or service requests through the ServiceNow system, escalating issues to higher tier teams if necessary.
- Handle calls with team members and utilize communication skills to troubleshoot and explain problems in an understandable manner.
- Administer and maintain accounts through the XP2 and Active Directory systems.
- Assist in imaging laptops for employee use.

Nicolas D. Swank
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10/22/2025
University of Michigan – College of Engineering
Ann Arbor, MI

Dear Hiring Team,

I am writing to express my interest in the Software Developer position within the Robotics Department at the University of Michigan. I recently graduated from Eastern Michigan University with a degree in Computer Science, and I've been building up my skills by taking on projects that let me apply what I'm learning in a hands-on way. This role feels like a good fit because it lines up with my background in programming, web development, and my drive to keep growing as a developer.

In college, I worked on a Pathfinding Visualizer project where our team built a map application. Like any map app, you could select two points, and it would calculate the path between them. We used different graph traversal algorithms to power it, and my part was writing Dijkstra's algorithm in TypeScript and making it work with the OpenStreetMap data. It was my first time applying algorithms to something people could actually use, and it taught me how to take a big idea and turn it into working code. Since then, I've been teaching myself full-stack development by building projects step by step. I created a To-do Task API in Node.js that uses JWT authentication so users can register, log in, and securely manage their tasks. I also built a recipe web app that connects to an external API, where I practiced organizing code using the MVC pattern. On top of that, I set up a SQL Server data warehouse, taking raw CSV files and transforming them into structured tables, giving me experience cleaning and working with real data. None of these projects were class assignments; I sought them out on my own because I wanted to learn the tools developers use every day.

I've always looked up to the University of Michigan for its innovation and research, and the chance to be part of the Robotics Department really stands out to me. It's exciting to think about working with people who are building technology that pushes boundaries and helps others learn. I'd love the opportunity to bring my skills into that environment and keep growing as part of the team.

Thank you for taking the time to review my application. I'd really appreciate the chance to talk more about the role.

Sincerely,
Nicolas Swank